

# Equity Quantitative Investment Strategies

From Financial Data to Systematic Investment Strategies

**Matéo Molinaro**

Quantitative Equity Researcher, VIA AM

September 10, 2026

## Matéo Molinaro

- Quantitative Equity Researcher.
- 3.5 YoE in Equity QIS (incl. internships): VIA AM, BNPP AM, Candriam, Lazard, Scientific Beta.
- Education: CPGE ENS D2, EDHEC, Dauphine, ENSAE
- Research focused on systematic equity strategies, statistical modelling and portfolio construction.

## Teaching objective

Transmit the theoretical and practical knowledge required to understand, implement and critically evaluate quantitative equity investment strategies.

## Course philosophy

- Mathematical foundations.
- Practical implementation.
- (Academic) Research discipline.
- Reproducible results.

# Course Overview

The course follows the complete quantitative equity research pipeline, from financial data to alpha generation, portfolio construction and performance evaluation.

## Data and Feature Engineering

- ➊ Introduction to Quantitative Equity Investing
- ➋ Financial Data Engineering
- ➌ Data Cleaning and Feature Engineering

## Research and Alpha Generation

- ➍ Signal Construction
- ➎ Statistical Models for Prediction
- ➏ Machine Learning for Prediction

## Portfolio and Implementation

- ➐ Portfolio Construction
- ➑ Portfolio Optimisation
- ➒ Backtesting Frameworks
- ➓ Performance Evaluation

## Final Project

Development, implementation and evaluation of an original quantitative equity investment strategy.

# Final Project

## Instruction

Choose a research paper related to the course, or formulate an original research question, and develop a quantitative equity investment strategy that addresses it.

## Expected research process

- 1 Define the investment hypothesis or research question
- 2 Acquire and prepare the required data
- 3 Construct features and investment signals
- 4 Develop and refine the modelling approach
- 5 Translate signals into portfolio weights
- 6 Backtest the strategy under realistic assumptions
- 7 Evaluate performance, robustness and limitations
- 8 Present and defend the research findings

# Final Project – Evaluation

## 1. Creativity and Originality – 3 pts

Original research question, investment hypothesis, features, modelling choices or portfolio design.

## 2. Strategy Improvement – 4 pts

Evolution from the initial strategy and Sharpe ratio through experiments, diagnostics and refinements to the final strategy.

## 3. Production-Grade Code – 8 pts

Clean, modular, reproducible and well-structured code following professional software-engineering practices.

## 4. Research Report – 8 pts

Crystal-clear presentation of the hypothesis, methodology, experiments, results, limitations and conclusions.

## 5. Live Presentation and Q&A – 4 pts

Clear presentation and ability to justify methodological choices and answer technical questions.

### Grading

Raw score: **27 pts**



Final grade: **/20**

### Research Process

Initial idea → first backtest → diagnosis → experiments → refinements → final strategy

# Final Project – Proposed Research Papers I

## Factor Investing, Timing and Portfolio Construction

- Darolles, Le Fol & Mero (2022), *Timing the Size Risk Premia*.
- van der Linden, Soebhag & van Vliet (2025), *Leveraging the Low-Volatility Effect*.
- Satchell et al. (2020), *Time-Series Variation in Factor Premia: The Influence of the Business Cycle*.
- van der Wal (2021), *Factor Timing Using a Markov-Switching Model*.
- Molinaro & Chaudron (2026), *Dynamic Allocation with Macro Factor-Mimicking Portfolios*.

## Machine Learning and Cross-Sectional Prediction

- Cong, Feng, He & Wang (2026), *Mosaics of Predictability*.
- Poh, Lim, Zohren & Roberts (2021), *Building Cross-Sectional Systematic Strategies by Learning to Rank*.
- Molinaro & Charbonnier (2026), *Predicting Analyst Performance: A Machine Learning Fund-of-Analysts Framework*.

# Final Project – Proposed Research Papers II

## NLP, LLMs and Alternative Data

- Zhang & Zhou (2026), *Large Language Models for Asset Pricing: Learning from Earnings Calls*.
- Molinaro & Montariol (2025), *Classifying US Firms into Trade Peace and Trade War Stocks: An NLP Analysis of Earnings Call Transcripts Approach*.
- Molinaro & Chuzeville (2026), *Do Earnings Call Transcripts Predict Post-Announcement Returns?*.

## Regimes and Market Structure

- Girnyk (2026), *Correlation Structures and Regime Shifts in Nordic Stock Markets*.

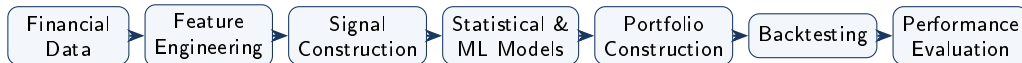
## Project Choice

Students may select one of the proposed papers or propose another research paper related to the course, subject to approval.

The objective is not only to reproduce the original results, but to **extend, challenge or improve** the proposed methodology and develop a complete quantitative investment strategy.

# Course Philosophy

The course follows the complete research workflow used to develop systematic equity investment strategies.



## Theory

- Economic intuition
- Mathematical formulation
- Derivations from first principles
- Statistical assumptions
- Limitations and biases

## Practice

- Python implementation
- Financial datasets
- Reproducible research
- Professional research architecture
- Final investment strategy

## Interview Focus

Concepts marked with ★ are particularly relevant for buy-side quantitative finance interviews.

# Learning Objectives

Upon successful completion of the course, students will be able to:

- Understand the complete quantitative equity research process
- Acquire, clean and structure financial datasets
- Engineer predictive financial features
- Derive and implement statistical learning models from first principles
- Construct systematic investment signals
- Build quantitative portfolios under realistic constraints
- Design robust backtesting frameworks
- Evaluate strategies using industry-standard metrics
- Identify statistical and backtesting biases
- Conduct reproducible quantitative research in Python

## Objective

Develop the theoretical and practical foundations required to conduct a complete quantitative equity research project.

# Technical Skills Developed

Students progressively develop a modern quantitative research environment comparable to those used in quantitative asset management.

## Quant Research

- Financial data
- Feature engineering
- Statistical modelling
- Machine learning
- Signal construction
- Portfolio optimisation
- Backtesting

## Python

- NumPy / pandas
- statsmodels
- scikit-learn
- Object-oriented programming (OOP)
- Modular code design
- Backtesting architecture

## Research Engineering

- IDE (PyCharm)
- Git and GitHub
- Virtual environments
- Dependency management (UV)
- Docker
- (Amazon S3 and EC2)

## Production Perspective

From isolated notebooks to **structured**, **modular**, **reproducible** and **documented** quantitative research.

# Who Is This Course For?

The course is designed to be accessible to students from diverse academic backgrounds.

## Finance and Economics

- Business schools
- Finance programmes
- Economics programmes
- Econometrics programmes

## Quantitative Disciplines

- Engineering schools
- Data science programmes
- Quantitative finance programmes
- Related quantitative disciplines

## A Common Quantitative Framework

Students may enter the course with different strengths in finance, mathematics or programming. The course progressively develops a common framework for quantitative equity research.

# Prerequisites

Basic mathematical and programming maturity is expected. No specialised quantitative finance background is required.

## Mathematical Foundations

Familiarity with basic concepts in:

- Linear algebra
- Probability
- Statistics
- Regression
- Optimisation

## Programming Foundations

Familiarity with basic concepts such as:

- Python (OOP)
- Variables and data types
- Conditions, loops and functions
- Basic data manipulation
- Basic numerical computation

## Progressive Learning

All required mathematical and computational tools are introduced, derived and developed progressively when needed.

## Chapter 1

# Introduction to Equity Quantitative Investment Strategies (EQIS)

# Chapter 1 – Introduction – Lecture Agenda

- 1 What is quantitative investing?
- 2 Active vs. passive investing
- 3 Discretionary vs. systematic vs. fundamental vs. quantitative investing
- 4 Alpha, beta and risk premia
- 5 Long-only vs. long/short strategies
- 6 Cross-sectional vs. time-series investing
- 7 Factor investing
- 8 Quantitative research workflow
- 9 Research reproducibility
- 10 Summary

## Main Objective

Understand what is an EQIS and what are its difficulties.

# What Is Quantitative Investing?

# What Is Quantitative Investing?

**Quantitative investing** is the use of **data**, **mathematical models** and **computational methods** to support investment decisions.

## Data

- Prices and returns
- Fundamentals (financial statements)
- Macroeconomic data
- Alternative data (news, 10K, 10Q, internet, satellite, ...)

## Models

- Model-free
- Statistics
- Econometrics
- Machine learning
- Optimisation

## Implementation

- Signal construction
- Portfolio construction
- Risk management
- Execution

## Key Idea

Quantitative investing transforms investment hypotheses into explicit, testable and reproducible decision rules ("trading" algorithms).

# What Is Quantitative Investing? – A Mathematical View

At each date  $t$ , the investor observes an **information set**  $\mathcal{I}_t$  containing all information available at that time.

A quantitative strategy defines an explicit mapping:

$$\boxed{\mathcal{I}_t \xrightarrow{f} w_t}$$

where

$$w_t = (w_{1,t} \quad \cdots \quad w_{N,t})^\top \in \mathbb{R}^N$$

is the vector of portfolio weights across  $N$  assets.

Equivalently,

$$\boxed{w_t = f(\mathcal{I}_t)}$$

## Point-in-Time Principle

$w_t$  may depend only on information available at time  $t$ : future information cannot enter  $\mathcal{I}_t$ .

# What Is Quantitative Investing? – From Weights to Returns

At date  $t$ , the strategy determines the portfolio weights

$$\mathbf{w}_t = (w_{1,t} \ \cdots \ w_{N,t})^\top \in \mathbb{R}^N.$$

The asset returns subsequently realised between  $t$  and  $t + 1$  are

$$\mathbf{R}_{t+1} = (R_{1,t+1} \ \cdots \ R_{N,t+1})^\top \in \mathbb{R}^N.$$

The portfolio return is therefore

$$R_{p,t+1} = \mathbf{w}_t^\top \mathbf{R}_{t+1} = \sum_{i=1}^N w_{i,t} R_{i,t+1}$$

## Timing

$\mathbf{w}_t$  is determined using information available at  $t$ , whereas  $\mathbf{R}_{t+1}$  is realised afterwards and is unknown at  $t$ .

# What Is Quantitative Investing? – What Makes a Strategy Quantitative?

A strategy becomes quantitative when its investment process is translated into **explicit and reproducible rules**.

## Investment Hypothesis

- What economic effect are we exploiting?
- Why should it exist?
- Why might it persist?

## Measurement

- Which data represent the hypothesis?
- How is the information transformed?
- How is the signal defined?

## Decision Rule

- Which securities are selected?
- How are positions sized?
- When is the portfolio rebalanced?

## Evaluation

- Does the strategy work out-of-sample?
- Is performance robust?
- Is it implementable after costs?

## Scientific Perspective

A quantitative strategy is a testable investment hypothesis, not simply a historical pattern.

# What Is Quantitative Investing? – From Data to Investment Decisions

Quantitative investing can be viewed as a sequence of transformations.



$$RD_t \xrightarrow{c} D_t \xrightarrow{h} X_t \xrightarrow{(f)} s_t \xrightarrow{g} w_t \xrightarrow{R_{t+1}} R_{p,t+1}$$

where:

- $RD_t$  = raw point-in-time dataset
- $D_t$  = cleaned point-in-time dataset
- $X_t$  = engineered feature matrix
- $(f)$  = optional statistical / ML model
- $s_t$  = investment signals
- $w_t$  = portfolio weights
- $R_{p,t+1}$  = subsequent portfolio return

## Course Perspective

Each transformation will be studied, derived, implemented and evaluated throughout the course.

# Active and Passive Investing

# Active versus passive investing

## Passive investing

The objective is to replicate the performance of a reference index or a predefined market exposure.

Examples:

- Market-capitalization-weighted index.
- Equally-weighted (EW) index.
- Global equity index.

## Active investing

**Traditional Asset Management:** The objective is typically to outperform a specified benchmark, subject to risk, exposure and investment constraints.

**Hedge Funds:** The objective is typically to generate attractive absolute risk-adjusted returns, with less emphasis on performance relative to a traditional benchmark.

Examples:

- Stock selection.
- Market timing.

Quantitative investing can be either passive or active. The distinction concerns the investment objective (outperformance), not the use of mathematical models.

# Active and Passive Investing – Active Management and the Benchmark

Let  $R_{p,t+1}$  denote the portfolio return and  $R_{b,t+1}$  the return of its benchmark.

The **active return** measures the portfolio's performance relative to the benchmark ★:

$$R_{p,t+1}^A = R_{p,t+1} - R_{b,t+1}$$

If the benchmark is the market portfolio,

$$R_{b,t+1} = R_{M,t+1},$$

then the active return becomes

$$R_{p,t+1}^A = R_{p,t+1} - R_{M,t+1}$$

# Fundamental, Quantitative, Discretionary and Systematic Investing

# Investment Styles – Two Different Dimensions

Investment approaches can be classified along **two distinct dimensions**.

## Information and Analysis

### Fundamental

Financial statements, business and industry analysis, valuation, management and competitive position.

### Quantitative

Structured datasets, statistical relationships, quantitative signals and mathematical / ML models.

## Decision Process

### Discretionary

Human judgement drives investment decisions. Rules may change case by case based on qualitative interpretation.

### Systematic

Investment decisions follow explicit and reproducible rules, typically implemented through algorithmic portfolio construction.

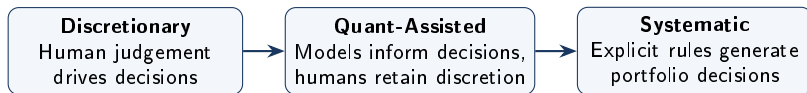
## Key Distinction

Fundamental  $\neq$  Discretionary

Quantitative  $\neq$  Systematic

# Investment Styles – From Discretionary to Systematic

Investment approaches differ primarily in **how investment decisions are generated**.



## Fundamental vs. Quantitative

**Fundamental data are not exclusive to discretionary investing.** Systematic quantitative strategies can use financial statements, valuation ratios, earnings revisions and other fundamental information.

## This Course

Equity Quantitative Investment Strategies focuses primarily on **systematic investing**: transforming data into explicit, reproducible portfolio decisions.

# Why systematic investing?

Systematic investing aims to transform an investment hypothesis into a reproducible decision process.

## ① Consistency

The same information is processed according to the same rules.

## ② Scalability

The same framework can be applied to many assets, markets and time periods.

## ③ Objectivity

Investment decisions are less dependent on individual judgement.

## ④ Researchability

Hypotheses can be tested statistically.

## ⑤ Reproducibility

Historical results can be reconstructed from the same inputs.

## Important

Systematic investing does not eliminate judgement. Judgement is transferred to the definition of the model, the data, the constraints and the research process.

# Alpha, Beta and Risk Premia

# Alpha, Beta and Risk Premia – Return Decomposition

A standard single-factor model decomposes asset returns as ★

$$R_{i,t} = \alpha_i + \beta_i F_t + \varepsilon_{i,t}$$

More generally, with  $K$  factors ★,

$$R_{i,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t} + \varepsilon_{i,t}$$

where

- $\alpha_i$  is the intercept,
- $\beta_{i,k}$  is the exposure of asset  $i$  to factor  $k$ ,
- $F_{k,t+1}$  is the return of factor  $k$ ,
- $\varepsilon_{i,t+1}$  is the residual return.

## Key Idea

Different asset-pricing models differ primarily in the factors  $F_{k,t+1}$  used to explain returns.

# Alpha, Beta and Risk Premia – From CAPM to Multi-Factor Models

The **CAPM** uses a single systematic risk factor: the market excess return ★.

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_M] - R_f)$$

with

$$\beta_i = \frac{\text{Cov}(R_i - R_f, R_M - R_f)}{\text{Var}(R_M - R_f)}$$

**APT** allows expected returns to depend on several systematic factors ★:

$$\mathbb{E}[R_i] - R_f = \sum_{k=1}^K \beta_{i,k} \lambda_k$$

where  $\lambda_k$  denotes the risk premium associated with factor  $k$ .

## Key Difference

CAPM relies on a **single systematic factor**, whereas APT allows **multiple systematic factors** to explain differences in expected returns.

## CAPM – Constant Risk-Free Rate

The CAPM beta is defined using excess returns:

$$\beta_i = \frac{\text{Cov}(R_i - R_f, R_M - R_f)}{\text{Var}(R_M - R_f)}$$

If  $R_f$  is constant over time, subtracting it does not affect covariances or variances:

$$\text{Cov}(R_i - R_f, R_M - R_f) = \text{Cov}(R_i, R_M),$$

$$\text{Var}(R_M - R_f) = \text{Var}(R_M).$$

Therefore,

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}$$

### Key Result

When  $R_f$  is constant, beta can equivalently be computed using raw returns or excess returns.

## CAPM – Empirical Estimation

In practice, the risk-free rate may vary through time. The CAPM is therefore estimated using **excess returns**:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{M,t} - R_{f,t}) + \varepsilon_{i,t}$$

for  $t = 1, \dots, T$ . Estimating by OLS gives

$$\hat{\beta}_i = \frac{\widehat{\text{Cov}}(R_i - R_f, R_M - R_f)}{\widehat{\text{Var}}(R_M - R_f)}$$

$$\hat{\alpha}_i = \overline{(R_i - R_f)} - \hat{\beta}_i \overline{(R_M - R_f)}$$

where  $\overline{(\cdot)}$  denotes the sample time-series average, e.g.

$$\overline{(R_i - R_f)} = \frac{1}{T} \sum_{t=1}^T (R_{i,t} - R_{f,t}).$$

**CAPM Test:** Under the CAPM,  $\alpha_i = 0$  in population; empirically, test

$$H_0 : \alpha_i = 0.$$

# Common Empirical Factor Models

Empirical asset-pricing models specify different sets of factors to explain the cross-section of returns.

| Model          | Factors                                                  | Number |
|----------------|----------------------------------------------------------|--------|
| CAPM           | Market                                                   | 1      |
| Fama–French 3  | Market, Size, Value                                      | 3      |
| Carhart        | Market, Size, Value, Momentum                            | 4      |
| Fama–French 5  | Market, Size, Value, Profitability, Investment           | 5      |
| FF5 + Momentum | Market, Size, Value, Profitability, Investment, Momentum | 6      |

For example, the Fama–French three-factor model is

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,M}MKT_t + \beta_{i,S}SMB_t + \beta_{i,V}HML_t + \varepsilon_{i,t}$$

## Interpretation

Adding factors changes what is considered “alpha”: a return unexplained by CAPM may be explained by exposure to size, value, momentum, or other systematic factors.

# Alpha, Beta and Risk Premia – Alpha – Definition

Consider the  $K$ -factor model:

$$R_{i,t+1} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t+1} + \varepsilon_{i,t+1}.$$

**Model parameters**

$$\alpha_i, \beta_{i,1}, \dots, \beta_{i,K}$$

Unknown parameters that can be estimated from historical data.

To interpret  $\alpha_i$  as an **expected unexplained return**, take expectations:

$$\mathbb{E}[R_{i,t+1}] = \alpha_i + \sum_{k=1}^K \beta_{i,k} \mathbb{E}[F_{k,t+1}] + \underbrace{\mathbb{E}[\varepsilon_{i,t+1}]}_{=0}.$$

Hence,  $\alpha_i$  captures the expected return not explained by the included factors.

**At time  $t$**

$$R_{i,t+1}, F_{k,t+1}, \varepsilon_{i,t+1}$$

Future random quantities, observed only after realization.

# Alpha, Beta and Risk Premia – Alpha – Interpretation

From the factor model,

$$\alpha_i = \mathbb{E}[R_{i,t+1}] - \sum_{k=1}^K \beta_{i,k} \mathbb{E}[F_{k,t+1}]$$

Thus, alpha is the portion of expected return not explained by the chosen factor exposures.

## Key Interpretation

Alpha is always defined relative to a model: changing the factor model can change the estimated alpha.

## Why Taking Expectations?

The shock  $\varepsilon_{i,t+1}$  is unobserved and varies across periods. Taking expectations and assuming  $\mathbb{E}[\varepsilon_{i,t+1}] = 0$  gives  $\alpha_i$  its economic interpretation as the **expected return unexplained by the factor model**. In empirical applications,  $\alpha_i$  is estimated jointly with the factor loadings  $\beta_{i,k}$ .

## Alpha, Beta and Risk Premia – Alpha – Empirical Estimation

In practice,  $\alpha_i$  and the factor exposures  $\beta_{i,k}$  are **unknown** and must be estimated from historical data.

For  $T$  historical observations, estimate the time-series regression

$$R_{i,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t} + \varepsilon_{i,t}, \quad t = 1, \dots, T$$

typically using **Ordinary Least Squares (OLS)** ★.

This produces

$$\hat{\alpha}_i, \quad \hat{\beta}_{i,1}, \dots, \hat{\beta}_{i,K},$$

where  $\hat{\alpha}_i$  is the estimated historical alpha.

The fitted residuals are then

$$\hat{\varepsilon}_{i,t} = R_{i,t} - \hat{\alpha}_i - \sum_{k=1}^K \hat{\beta}_{i,k} F_{k,t}.$$

# Alpha, Beta and Risk Premia – Beta

Consider the single-factor market model:

$$R_{i,t+1} = \alpha_i + \beta_i R_{M,t+1} + \varepsilon_{i,t+1}.$$

Assume the idiosyncratic component is uncorrelated with the market (see next slides) ★:

$$\text{Cov}(R_{M,t+1}, \varepsilon_{i,t+1}) = 0.$$

Taking the covariance with the market return:

$$\begin{aligned}\text{Cov}(R_i, R_M) &= \text{Cov}(\alpha_i + \beta_i R_M + \varepsilon_i, R_M) \\ &= \underbrace{\text{Cov}(\alpha_i, R_M)}_{=0} + \beta_i \text{Var}(R_M) + \underbrace{\text{Cov}(\varepsilon_i, R_M)}_{=0} \\ &= \beta_i \text{Var}(R_M).\end{aligned}$$

Therefore ★,

$$\boxed{\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}}$$

# Alpha, Beta and Risk Premia – Beta – Interpretation

Beta measures the **sensitivity of asset  $i$ 's return** to the contemporaneous market return ★:

$$R_{i,t} = \alpha_i + \beta_i R_{M,t} + \varepsilon_{i,t},$$

Market-related component =  $\beta_i R_{M,t}$

| Beta              | Interpretation                             |
|-------------------|--------------------------------------------|
| $\beta_i < 0$     | Asset tends to move opposite to the market |
| $\beta_i = 0$     | No linear exposure to market movements     |
| $0 < \beta_i < 1$ | Less sensitive than the market             |
| $\beta_i = 1$     | Same sensitivity as the market             |
| $\beta_i > 1$     | More sensitive than the market             |

## Example

If  $\beta_i = 1.5$ , a +1% market return is associated, on average, with approximately a +1.5% contribution to asset  $i$ 's return through its market exposure, holding the other components fixed.

# Alpha, Beta and Risk Premia – Beta – Estimation

In the population, the market beta is defined as

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}$$

where  $\text{Cov}(R_i, R_M)$  and  $\text{Var}(R_M)$  are **population moments** and are generally unknown.

In practice, we observe a sample of  $T$  historical returns and estimate

$$\hat{\beta}_i = \frac{\widehat{\text{Cov}}(R_i, R_M)}{\widehat{\text{Var}}(R_M)}$$

## Population vs. Sample

$\beta_i$  is the true, unknown population beta, while  $\hat{\beta}_i$  is its estimate obtained from a finite sample of returns.

## Alpha, Beta and Risk Premia – Beta – Sample Estimate

Using the sample covariance and variance,

$$\hat{\beta}_i = \frac{\frac{1}{T-1} \sum_{t=1}^T (R_{M,t} - \bar{R}_M)(R_{i,t} - \bar{R}_i)}{\frac{1}{T-1} \sum_{t=1}^T (R_{M,t} - \bar{R}_M)^2}.$$

The common factor  $1/(T-1)$  cancels:

$$\hat{\beta}_i = \frac{\sum_{t=1}^T (R_{M,t} - \bar{R}_M)(R_{i,t} - \bar{R}_i)}{\sum_{t=1}^T (R_{M,t} - \bar{R}_M)^2}$$

### OLS Equivalence

This expression is exactly the OLS slope estimator from the regression

$$R_{i,t} = \alpha_i + \beta_i R_{M,t} + \varepsilon_{i,t}.$$

Thus, in a single-factor regression with an intercept, the sample covariance/variance approach and OLS give the same  $\hat{\beta}_i$  ★.

# Alpha, Beta and Risk Premia – Beta – Population Model

For asset  $i$  at date  $t$ , consider the single-factor model

$$R_{i,t} = \alpha_i + \beta_i R_{M,t} + \varepsilon_{i,t}$$

where

$$R_{i,t}, R_{M,t}, \varepsilon_{i,t} \in \mathbb{R}, \quad \alpha_i, \beta_i \in \mathbb{R}.$$

| Object              | Interpretation                          |
|---------------------|-----------------------------------------|
| $R_{i,t}$           | Return of asset $i$ at date $t$         |
| $R_{M,t}$           | Market return at date $t$               |
| $\alpha_i$          | Population intercept for asset $i$      |
| $\beta_i$           | Population market exposure of asset $i$ |
| $\varepsilon_{i,t}$ | Idiosyncratic residual at date $t$      |

The residual realization is therefore

$$\varepsilon_{i,t} = R_{i,t} - \alpha_i - \beta_i R_{M,t}.$$

# Alpha, Beta and Risk Premia – Beta – Population vs. Sample

In the **population**,  $(\alpha_i, \beta_i)$  are defined by the linear projection

$$(\alpha_i, \beta_i) = \arg \min_{\alpha, \beta} \mathbb{E} \left[ (R_{i,t} - \alpha - \beta R_{M,t})^2 \right]$$

where  $(R_{i,t}, R_{M,t})$  is treated as a random vector and the expectation averages over its population distribution.

In a **sample** of  $T$  observed returns, OLS instead estimates

$$(\hat{\alpha}_i, \hat{\beta}_i) = \arg \min_{\alpha, \beta} \sum_{t=1}^T (R_{i,t} - \alpha - \beta R_{M,t})^2$$

## Notation

$(\alpha_i, \beta_i)$  are population coefficients, whereas  $(\hat{\alpha}_i, \hat{\beta}_i)$  are estimates obtained from a finite sample. The orthogonality derivation that follows concerns the **population linear projection**.

# Alpha, Beta and Risk Premia – Beta – Population Objective

For a fixed asset  $i$ , define the population objective function

$$Q_i(\alpha, \beta) = \mathbb{E} \left[ (R_{i,t} - \alpha - \beta R_{M,t})^2 \right]$$

The optimization variables are the scalars

$$\alpha, \beta \in \mathbb{R},$$

and the minimizing values are the asset-specific population coefficients

$$(\alpha_i, \beta_i) = \arg \min_{\alpha, \beta} Q_i(\alpha, \beta).$$

At the optimum, the first-order conditions are

$$\left. \frac{\partial Q_i}{\partial \alpha} \right|_{(\alpha_i, \beta_i)} = 0 \quad \text{and} \quad \left. \frac{\partial Q_i}{\partial \beta} \right|_{(\alpha_i, \beta_i)} = 0.$$

## Alpha, Beta and Risk Premia – Beta – FOC for $\alpha_i$

Starting from

$$Q_i(\alpha, \beta) = \mathbb{E} [(R_{i,t} - \alpha - \beta R_{M,t})^2],$$

differentiate with respect to  $\alpha$ :

$$\begin{aligned}\frac{\partial Q_i}{\partial \alpha} &= \mathbb{E} \left[ \frac{\partial}{\partial \alpha} (R_{i,t} - \alpha - \beta R_{M,t})^2 \right] \\ &= \mathbb{E} \left[ 2(R_{i,t} - \alpha - \beta R_{M,t}) \frac{\partial}{\partial \alpha} (R_{i,t} - \alpha - \beta R_{M,t}) \right] \\ &= \mathbb{E} [2(R_{i,t} - \alpha - \beta R_{M,t})(-1)] \\ &= -2\mathbb{E} [R_{i,t} - \alpha - \beta R_{M,t}].\end{aligned}$$

## Alpha, Beta and Risk Premia – Beta – FOC for $\alpha_i$

Using linearity of expectation,

$$\begin{aligned}\frac{\partial Q_i}{\partial \alpha} &= -2\mathbb{E}[R_{i,t} - \alpha - \beta R_{M,t}] \\ &= -2(\mathbb{E}[R_{i,t}] - \mathbb{E}[\alpha] - \mathbb{E}[\beta R_{M,t}]) \\ &= -2(\mathbb{E}[R_{i,t}] - \alpha - \beta \mathbb{E}[R_{M,t}]),\end{aligned}$$

since  $\alpha, \beta \in \mathbb{R}$  are constants.

Evaluating the first-order condition at  $(\alpha_i, \beta_i)$  gives

$$0 = -2(\mathbb{E}[R_{i,t}] - \alpha_i - \beta_i \mathbb{E}[R_{M,t}]).$$

Hence,

$$\boxed{\mathbb{E}[R_{i,t}] - \alpha_i - \beta_i \mathbb{E}[R_{M,t}] = 0}.$$

# Alpha, Beta and Risk Premia – Beta – Zero-Mean Residual

Recall the population residual at date  $t$ :

$$\varepsilon_{i,t} = R_{i,t} - \alpha_i - \beta_i R_{M,t}.$$

Taking expectations,

$$\begin{aligned}\mathbb{E}[\varepsilon_{i,t}] &= \mathbb{E}[R_{i,t} - \alpha_i - \beta_i R_{M,t}] \\ &= \mathbb{E}[R_{i,t}] - \alpha_i - \beta_i \mathbb{E}[R_{M,t}].\end{aligned}$$

But the first-order condition for  $\alpha_i$  gave

$$\mathbb{E}[R_{i,t}] - \alpha_i - \beta_i \mathbb{E}[R_{M,t}] = 0.$$

Therefore,

$$\boxed{\mathbb{E}[\varepsilon_{i,t}] = 0}.$$

## First Population Orthogonality Condition

Because the linear projection includes an intercept, its population residual has zero mean.

## Alpha, Beta and Risk Premia – Beta – FOC for $\beta_i$

Now differentiate

$$Q_i(\alpha, \beta) = \mathbb{E}[(R_{i,t} - \alpha - \beta R_{M,t})^2]$$

with respect to  $\beta$ :

$$\begin{aligned}\frac{\partial Q_i}{\partial \beta} &= \mathbb{E} \left[ 2(R_{i,t} - \alpha - \beta R_{M,t}) \frac{\partial}{\partial \beta} (R_{i,t} - \alpha - \beta R_{M,t}) \right] \\ &= \mathbb{E} [2(R_{i,t} - \alpha - \beta R_{M,t})(-R_{M,t})] \\ &= -2\mathbb{E} [R_{M,t}(R_{i,t} - \alpha - \beta R_{M,t})].\end{aligned}$$

At the optimum  $(\alpha_i, \beta_i)$ ,

$$0 = -2\mathbb{E} [R_{M,t}(R_{i,t} - \alpha_i - \beta_i R_{M,t})].$$

## Alpha, Beta and Risk Premia – Beta – Orthogonality Condition

From the first-order condition for  $\beta_i$ ,

$$0 = -2\mathbb{E}[R_{M,t}(R_{i,t} - \alpha_i - \beta_i R_{M,t})].$$

Recall that

$$\varepsilon_{i,t} = R_{i,t} - \alpha_i - \beta_i R_{M,t}.$$

Substituting,

$$0 = -2\mathbb{E}[R_{M,t}\varepsilon_{i,t}].$$

Therefore,

$$\boxed{\mathbb{E}[R_{M,t}\varepsilon_{i,t}] = 0}.$$

### Second Population Orthogonality Condition

The population residual is orthogonal to the market regressor: their expected product is zero.

# Alpha, Beta and Risk Premia – Beta – Residual Orthogonality

Recall

$$\text{Cov}(R_{M,t}, \varepsilon_{i,t}) = \mathbb{E}[R_{M,t}\varepsilon_{i,t}] - \mathbb{E}[R_{M,t}]\mathbb{E}[\varepsilon_{i,t}].$$

The two population first-order conditions gave

$$\boxed{\mathbb{E}[\varepsilon_{i,t}] = 0} \quad \text{and} \quad \boxed{\mathbb{E}[R_{M,t}\varepsilon_{i,t}] = 0}.$$

Therefore,

$$\text{Cov}(R_{M,t}, \varepsilon_{i,t}) = \mathbb{E}[R_{M,t}\varepsilon_{i,t}] - \mathbb{E}[R_{M,t}]\mathbb{E}[\varepsilon_{i,t}] = 0 - \mathbb{E}[R_{M,t}] \times 0 = \boxed{0}.$$

Hence,

$$\boxed{\text{Cov}(R_{M,t}, \varepsilon_{i,t}) = 0}.$$

## Interpretation

The population linear projection chooses  $\alpha_i$  and  $\beta_i$  such that no remaining **linear market exposure** is contained in the residual.

## Risk premia

A risk premium is the expected compensation associated with exposure to a particular source of risk ★.

Let  $F_{k,t+1}$  denote the return of factor  $k$ .

A factor premium is

$$\lambda_k = \mathbb{E}[F_{k,t+1}].$$

The expected return of asset  $i$  is

$$\mathbb{E}[R_{i,t+1}] = \alpha_i + \sum_{k=1}^K \beta_{i,k} \lambda_k.$$

Examples of commonly studied equity risk premia:

- Market premium.
- Value premium.
- Momentum premium.

# Alpha versus risk premia

## Risk premium

Expected return associated with exposure to a systematic factor.

$$\mathbb{E}[R_i] = \beta_i \lambda$$

The return is attributed to exposure to a systematic source of risk.

## Alpha

Expected return unexplained by the factors included in the model.

$$\alpha_i = \mathbb{E}[R_i] - \sum_k \beta_{i,k} \lambda_k$$

Alpha is therefore model-dependent.

## Key question

Is the return generated by a strategy compensation for systematic exposures, or is it unexplained by the chosen model?

## Long-Only and Long/Short Strategies

# Long-Only Investing

A long-only portfolio satisfies

$$w_{i,t} \geq 0 \quad \forall i, \quad \sum_{i=1}^N w_{i,t} = 1$$

where the second condition corresponds to a **fully invested** portfolio.

Given the weight vector  $w_t = (w_{1,t}, \dots, w_{N,t})^\top \in \mathbb{R}^N$ , the subsequent portfolio return is

$$R_{p,t+1} = w_t^\top R_{t+1} = \sum_{i=1}^N w_{i,t} R_{i,t+1}$$

## Interpretation

The investor allocates non-negative portfolio weights to assets and does not take explicit short positions.

## Typical applications

- Equity mutual funds
- Long-only factor portfolios
- Index tracking

# Long/Short Investing – Net and Gross Exposure

A long/short portfolio allows both positive and negative weights:

$$w_{i,t} \in \mathbb{R}$$

Two quantities characterize the portfolio's exposure:

## Net Exposure

$$\text{Net}_t = \sum_{i=1}^N w_{i,t}$$

Long exposure minus short exposure.

## Gross Exposure

$$\text{Gross}_t = \sum_{i=1}^N |w_{i,t}|$$

Total absolute capital exposure.

## Interpretation

Net exposure measures the portfolio's directional capital exposure, while gross exposure measures the total size of its long and short positions.

## Long/Short Investing – Leverage

Leverage measures the portfolio's **total absolute exposure**.

If  $w_{i,t}$  denotes the portfolio weight of asset  $i$ , leverage is

$$L_t = \sum_{i=1}^N |w_{i,t}|$$

Therefore, leverage corresponds directly to the portfolio's **gross exposure**:

$$L_t = \text{Gross}_t$$

### Example 1

100% long

$$L_t = |1| = 1$$

### Example 2

100% long / 100% short

$$L_t = |1| + |-1| = 2$$

**Key idea:** A portfolio can have zero net exposure while having substantial leverage. For example, a 100% long / 100% short portfolio has  $\text{Net}_t = 0$  but  $L_t = 2$ .

## Long/Short Investing – Dollar Neutrality

A portfolio is **dollar neutral** when its long and short dollar exposures offset each other:

$$\text{Net}_t = \sum_{i=1}^N w_{i,t} = 0$$

For example, a portfolio with

$$\sum_{i:w_{i,t}>0} w_{i,t} = +1 \quad \text{and} \quad \sum_{i:w_{i,t}<0} w_{i,t} = -1$$

has

$$\text{Net}_t = 0, \quad \text{Gross}_t = 2.$$

# Long/Short Investing – Dollar vs. Beta Neutrality

**Dollar neutrality** requires zero net exposure:

$$\sum_{i=1}^N w_{i,t} = 0$$

However, assets can have different market betas. The portfolio beta is

$$\beta_{p,t} = \sum_{i=1}^N w_{i,t} \beta_{i,t} = \mathbf{w}_t^\top \boldsymbol{\beta}_t$$

where

$$\mathbf{w}_t, \boldsymbol{\beta}_t \in \mathbb{R}^N, \quad \beta_{p,t} \in \mathbb{R}.$$

Therefore,

$$\sum_{i=1}^N w_{i,t} = 0 \quad \not\Rightarrow \quad \sum_{i=1}^N w_{i,t} \beta_{i,t} = 0.$$

# Cross-Sectional and Time-Series Investing

# Cross-Sectional Investing

Cross-sectional investing compares **different assets at the same point in time**.

Let

$$x_{i,t} \in \mathbb{R}$$

denote a characteristic of asset  $i$  observed at time  $t$ .

## Examples of characteristics:

- Valuation ratio
- Profitability
- Earnings revision
- Momentum score
- Volatility
- Quality score

At each date  $t$ , the characteristic is compared **across stocks** in the investment universe.

## Objective

Identify assets that are expected to outperform other assets in the same investment universe.

# Cross-Sectional Investing – From Characteristics to Rankings

At time  $t$ , suppose the investment universe contains  $N$  assets:

$$\mathcal{U}_t = \{1, \dots, N\}.$$

Each asset receives a characteristic  $x_{i,t}$ , which can be transformed into an investment signal:

$$s_{i,t} = f(x_{i,t})$$

The assets are then compared and ranked **cross-sectionally**.  
For example, if a higher signal is considered more attractive,

$$s_{i,t} > s_{j,t} \implies \text{asset } i \text{ is ranked above asset } j.$$

Portfolio weights are then obtained from the signals or their ranks:

$$w_{i,t} = g(s_{i,t})$$

## Key Idea

The strategy asks: *which stocks are relatively more attractive than the others today?*

## Cross-Sectional Investing – Decile Portfolio

A common implementation is to sort stocks by their signal and form **equal-weighted decile portfolios**.

Consider a long/short strategy that:

- buys the **top decile** of stocks,
- shorts the **bottom decile**,
- assigns zero weight to all remaining stocks.

The portfolio weights are

$$w_{i,t} = \begin{cases} +\frac{1}{N_H}, & \text{if } i \text{ belongs to the top decile,} \\ -\frac{1}{N_L}, & \text{if } i \text{ belongs to the bottom decile,} \\ 0, & \text{otherwise,} \end{cases}$$

where  $N_H$  and  $N_L$  denote the numbers of stocks in the top and bottom deciles.

# Time-Series Investing

Time-series investing studies an asset using its **own historical information**.

Let

$$R_{i,t} \in \mathbb{R}$$

denote the return of asset  $i$  at time  $t$ .

A time-series signal can be constructed from past observations:

$$s_{i,t} = f(R_{i,t}, R_{i,t-1}, \dots, R_{i,t-L+1})$$

where  $L$  denotes the historical lookback window.

**Examples of time-series strategies:**

- Trend following
- Mean reversion
- Market timing
- Time-series momentum
- Volatility timing
- Regime signals

## Objective

Determine whether the future return of an asset is related to information contained in its own historical behaviour.

# Time-Series Investing – From History to Position

Unlike cross-sectional investing, an asset does not need to be ranked relative to other assets. Its own history determines its signal:

$$(R_{i,t}, R_{i,t-1}, \dots) \longrightarrow s_{i,t} \longrightarrow w_{i,t}$$

For example, a simple trend signal can use the cumulative return over the previous  $L$  periods:

$$s_{i,t} = \prod_{\tau=t-L+1}^t (1 + R_{i,\tau}) - 1.$$

The position can then depend on the sign of the signal:

$$w_{i,t} = \begin{cases} +1, & s_{i,t} > 0, \\ -1, & s_{i,t} < 0. \end{cases}$$

## Key Idea

The strategy asks: *based on this asset's own history, should I be long or short the asset today?*

# Cross-Sectional vs. Time-Series Investing

| Cross-Sectional                            | Time-Series                               |
|--------------------------------------------|-------------------------------------------|
| Compare different assets at date $t$       | Study an asset through time               |
| Relative signal / ranking                  | Absolute signal                           |
| “Which stocks are more attractive?”        | “Should I be long or short this asset?”   |
| Often used for stock selection             | Often used for trend / market timing      |
| Signal depends on relative characteristics | Signal depends on the asset’s own history |

## Fundamental Distinction

Cross-sectional strategies exploit **differences across assets**, whereas time-series strategies exploit **patterns through time**. The distinction affects signal construction, portfolio construction, and statistical analysis.

# Factor Investing

# What Is Factor Investing?

Factor investing targets **systematic characteristics** associated with differences in expected returns or risk across assets.

**Common equity factors include:**

- Market
- Value
- Momentum
- Size
- Quality
- Low volatility

A factor is often represented empirically by a **factor portfolio** whose return is denoted

$$F_{k,t+1}$$

for factor  $k$  over period  $t$  to  $t + 1$ .

## Key Idea

Factor investing seeks systematic exposure to common sources of return or risk rather than relying only on individual stock selection.

# Factors and Expected Returns

In a multi-factor model,

$$R_{i,t+1} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t+1} + \varepsilon_{i,t+1}$$

Taking expectations,

$$\mathbb{E}[R_i] = \alpha_i + \sum_{k=1}^K \beta_{i,k} \lambda_k, \quad \lambda_k = \mathbb{E}[F_k].$$

**Factor Exposure**

$$\beta_{i,k}$$

Sensitivity of asset  $i$  to factor  $k$ .

**Factor Premium**

$$\lambda_k$$

Expected return of factor  $k$ .

# Estimating Factor Exposures and Risk Premia

The factor model contains two distinct quantities:

$$\boxed{\beta_{i,k}} \quad \text{and} \quad \boxed{\lambda_k}.$$

They answer different empirical questions.

## Time-Series Dimension

$$R_{i,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t} + \varepsilon_{i,t}$$

Using observations through **time** for asset  $i$  allows us to estimate its factor exposures

$$\boxed{\hat{\beta}_{i,k}}.$$

**Key distinction:**  $\beta_{i,k}$  measures how much asset  $i$  is exposed to factor  $k$ ;  $\lambda_k$  measures how much expected return is associated with that exposure.

## Cross-Sectional Dimension

$$\mathbb{E}[R_i] = \alpha_i + \sum_{k=1}^K \beta_{i,k} \lambda_k$$

Comparing expected returns **across assets** allows us to estimate the factor premia

$$\boxed{\hat{\lambda}_k}.$$

# Fama–MacBeth Regression – Step 1

Fama–MacBeth estimates factor risk premia in two stages.

## Step 1 – Estimate Factor Exposures

For each asset  $i$ , estimate the time-series factor model

$$R_{i,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t} + \varepsilon_{i,t}$$

for

$$t = 1, \dots, T.$$

This provides the estimated factor exposures

$$\hat{\beta}_{i,1}, \dots, \hat{\beta}_{i,K}$$

for each asset  $i$ .

**Time-series variation  $\implies$  estimated factor sensitivities**

## Fama–MacBeth Regression – Step 2

The estimated factor exposures are now used to estimate **factor risk premia**.

### Step 2 – Estimate Factor Risk Premia

At each date  $t$ , run a cross-sectional regression across assets:

$$R_{i,t} = \gamma_{0,t} + \sum_{k=1}^K \hat{\beta}_{i,k} \lambda_{k,t} + u_{i,t}$$

for

$$i = 1, \dots, N.$$

This provides an estimated factor premium  $\hat{\lambda}_{k,t}$  for each factor  $k$  at each date  $t$ .

Average these estimates through time:

$$\hat{\lambda}_k = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_{k,t}$$

**Cross-sectional variation  $\Rightarrow$  estimated factor risk premia**

# Factor Return vs Factor Risk Premium

Although both refer to factor  $k$ ,  $F_{k,t}$  and  $\lambda_{k,t}$  represent **different quantities**.

## Factor Return

$$F_{k,t}$$

Realized return of factor  $k$  at date  $t$ .  
It enters the **time-series regression**:

$$R_{i,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t} + \varepsilon_{i,t}.$$

**Role:** used to estimate the factor exposure  $\beta_{i,k}$ .

## Factor Risk Premium

$$\lambda_{k,t}$$

Cross-sectional return associated with one unit of exposure to factor  $k$  at date  $t$ .  
It enters the **cross-sectional regression**:

$$R_{i,t} = \gamma_{0,t} + \sum_{k=1}^K \beta_{i,k} \lambda_{k,t} + u_{i,t}.$$

**Role:** measures whether exposure to factor  $k$  is rewarded in the cross-section.

$$F_{k,t} \longrightarrow \hat{\beta}_{i,k} \longrightarrow \hat{\lambda}_{k,t}$$

**Key distinction:** factor returns help estimate **exposures**; factor exposures help estimate **risk premia**.

# Factor Returns vs Factor Risk Premia – Setup I

Consider  $N$  assets and  $K$  priced factors.

## Asset Returns

$$R_t = \begin{bmatrix} R_{1,t} \\ \vdots \\ R_{N,t} \end{bmatrix} \in \mathbb{R}^N$$

where  $R_{i,t}$  is the return of asset  $i$  at date  $t$ .

## Factor Exposure Matrix

$$B = \begin{bmatrix} \beta_{1,1} & \cdots & \beta_{1,K} \\ \vdots & \ddots & \vdots \\ \beta_{N,1} & \cdots & \beta_{N,K} \end{bmatrix} \in \mathbb{R}^{N \times K}$$

where  $\beta_{i,k}$  measures the exposure of asset  $i$  to factor  $k$ .

Rows of  $B$  correspond to assets; columns correspond to factors.

# Factor Returns vs Factor Risk Premia – Setup II

## Factor Risk Premia

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_K \end{bmatrix} \in \mathbb{R}^K$$

where  $\lambda_k$  is the expected return compensation associated with one unit of exposure to factor  $k$ .

The cross-sectional expected-return relation is

$$\mathbb{E}[R] = \lambda_0 \mathbf{1}_N + \mathbf{B}\boldsymbol{\lambda}$$

with

$$\lambda_0 \in \mathbb{R}, \quad \mathbf{1}_N = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^N.$$

## Time Indexing – What Depends on $t$ ?

Some objects vary at each date, while others are initially assumed **constant over the estimation period**.

| Object          | Dimension                 | Interpretation                                                |
|-----------------|---------------------------|---------------------------------------------------------------|
| $R_t$           | $\mathbb{R}^N$            | Realized cross-section of asset returns at date $t$           |
| $B$             | $\mathbb{R}^{N \times K}$ | Factor exposures, assumed constant over the estimation period |
| $\lambda_t$     | $\mathbb{R}^K$            | Cross-sectional factor prices at date $t$                     |
| $\lambda$       | $\mathbb{R}^K$            | Unconditional / average factor risk premia                    |
| $\mathbb{E}[R]$ | $\mathbb{R}^N$            | Unconditional expected asset returns                          |

Thus, we distinguish the **unconditional pricing relation**

$$\mathbb{E}[R] = \lambda_0 1_N + B\lambda$$

from the **period- $t$  cross-sectional relation**

$$R_t = \gamma_{0,t} 1_N + B\lambda_t + u_t$$

**Later extension:** factor exposures may also vary through time, in which case  $B$  becomes  $B_t$ .

# Factor-Mimicking Portfolios

Construct  $K$  factor-mimicking portfolios using the  $N$  assets.

## Portfolio Weight Matrix

$$W = \begin{bmatrix} | & & | \\ w_1 & \cdots & w_K \\ | & & | \end{bmatrix} \in \mathbb{R}^{N \times K}, \quad w_k \in \mathbb{R}^N$$

where  $w_k$  contains the weights of factor portfolio  $k$ .

## Factor Portfolio Returns

The vector of realized factor returns is

$$F_t = W^\top R_t \in \mathbb{R}^K.$$

For factor  $k$  individually,

$$F_{k,t} = w_k^\top R_t = \sum_{i=1}^N w_{i,k} R_{i,t} \in \mathbb{R}.$$

**Interpretation:**  $F_{k,t}$  is the realized return at date  $t$  of the portfolio constructed to mimic factor  $k$ .

# Expected Return of Factor-Mimicking Portfolios

Starting from

$$F_t = W^\top R_t,$$

take expectations:

$$\mathbb{E}[F] = W^\top \mathbb{E}[R].$$

Using

$$\mathbb{E}[R] = \lambda_0 1_N + B\lambda,$$

we obtain

$$\mathbb{E}[F] = W^\top (\lambda_0 1_N + B\lambda).$$

Hence,

$$\boxed{\mathbb{E}[F] = \lambda_0 W^\top 1_N + W^\top B\lambda.}$$

This equation determines when the expected return of a factor portfolio equals the corresponding factor risk premium.

# When Does $\mathbb{E}[F] = \lambda$ ? – Condition 1

Start from

$$\mathbb{E}[F] = \lambda_0 W^\top 1_N + W^\top B \lambda.$$

## Condition 1 – Zero-Investment Portfolios

Each factor portfolio is long/short with zero net exposure:

$$W^\top 1_N = 0_K \in \mathbb{R}^K$$

Therefore, the intercept component vanishes:

$$\lambda_0 W^\top 1_N = 0_K.$$

Hence,

$$\mathbb{E}[F] = W^\top B \lambda.$$

**Interpretation:** Because each factor portfolio has zero net exposure, the common intercept  $\lambda_0$  does not contribute to its expected return.

## When Does $\mathbb{E}[F] = \lambda$ ? – Condition 2

After imposing zero investment,

$$\mathbb{E}[F] = W^\top B \lambda.$$

### Condition 2 – Unit-Exposure Normalization

Require

$$W^\top B = I_K, \quad I_K \in \mathbb{R}^{K \times K}$$

where  $I_K$  is the identity matrix.

This means that factor portfolio  $k$  has

$$\beta_{F_k, k} = 1, \quad \beta_{F_k, j} = 0 \quad \forall j \neq k$$

i.e. unit exposure to factor  $k$  and zero exposure to all other priced factors.

Therefore,

$$\mathbb{E}[F] = I_K \lambda = \lambda,$$

hence

## Factor Return and Factor Premium – One-Factor Case

Suppose expected returns satisfy

$$\mathbb{E}[R_i] = \lambda_0 + \beta_i \lambda.$$

Construct a factor portfolio

$$F_t = \mathbf{w}^\top \mathbf{R}_t, \quad \mathbf{w} \in \mathbb{R}^N.$$

Then

$$\mathbb{E}[F] = \mathbf{w}^\top \mathbb{E}[\mathbf{R}].$$

Substituting the pricing relation,

$$\mathbb{E}[F] = \lambda_0 \mathbf{w}^\top \mathbf{1}_N + \lambda \mathbf{w}^\top \boldsymbol{\beta},$$

where

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_N \end{bmatrix} \in \mathbb{R}^N.$$

## When Are Expected Factor Returns and Premia Different?

Suppose the factor portfolios are zero investment,

$$W^\top 1_N = 0_K,$$

but

$$W^\top B = A, \quad A \neq I_K.$$

Here,

$$A \in \mathbb{R}^{K \times K}$$

contains the actual exposures of the factor portfolios to the priced factors.

Then

$$\mathbb{E}[F] = A\lambda.$$

If  $A$  is invertible,

$$\lambda = A^{-1}\mathbb{E}[F].$$

Therefore, a portfolio called “Value” “Momentum” etc. does not automatically have expected return equal to the

## When They Differ – Numerical Example

Consider a one-factor model.

Suppose a Value factor portfolio has exposure

$$\beta_F = 1.4$$

to the underlying priced Value factor.

Its expected return satisfies

$$\mathbb{E}[F] = \beta_F \lambda.$$

Suppose empirically

$$\mathbb{E}[F] = 7\%.$$

Then

$$7\% = 1.4 \lambda,$$

and therefore

$$\lambda = \frac{7\%}{1.4} = 5\%.$$

## Factor Return vs Factor Premium – Period by Period

The previous equality concerned **expected returns**.

Now consider the period- $t$  cross-sectional relation

$$R_t = \gamma_{0,t} \mathbf{1}_N + B \boldsymbol{\lambda}_t + u_t$$

where

$$\boldsymbol{\lambda}_t \in \mathbb{R}^K$$

is the vector of period- $t$  cross-sectional factor premia, and

$$u_t \in \mathbb{R}^N$$

is the vector of cross-sectional residual returns.

Because

$$F_t = W^\top R_t,$$

we obtain

$$F_t = \gamma_{0,t} W^\top \mathbf{1}_N + W^\top B \boldsymbol{\lambda}_t + W^\top u_t.$$

## When Does $F_t = \lambda_t$ ?

Assume again that the factor portfolios satisfy

$$W^\top 1_N = 0_K$$

and

$$W^\top B = I_K.$$

Then

$$F_t = \lambda_t + W^\top u_t.$$

Therefore,

$$F_t = \lambda_t + W^\top u_t.$$

Period-by-period equality requires the additional condition

$$W^\top u_t = 0_K.$$

Hence,

# Expected Equality vs Period-by-Period Equality

Two different statements must be distinguished.

## Expected Equality

$$\mathbb{E}[F] = \lambda$$

Requires

$$W^T 1_N = 0_K$$

and

$$W^T B = I_K.$$

## Period-by-Period Equality

$$F_t = \lambda_t$$

Requires the previous conditions plus

$$W^T u_t = 0_K.$$

Therefore,

$$\mathbb{E}[F_k] = \lambda_k \not\Rightarrow F_{k,t} = \lambda_{k,t} \quad \forall t.$$

## Factor Return vs Factor Risk Premium – Summary

| Object            | Meaning                                 | Definition / Estimation                |
|-------------------|-----------------------------------------|----------------------------------------|
| $F_{k,t}$         | Realized return of factor portfolio $k$ | $w_k^\top R_t$                         |
| $\mathbb{E}[F_k]$ | Expected return of factor portfolio     | Expected value of $F_{k,t}$            |
| $\lambda_{k,t}$   | Period- $t$ price of factor exposure    | Cross-sectional regression coefficient |
| $\lambda_k$       | Average factor risk premium             | Expected / average price of exposure   |

Under zero-investment and unit-exposure normalization,

$$\mathbb{E}[F_k] = \lambda_k.$$

But period by period,

$$F_{k,t} = \lambda_{k,t} + w_k^\top u_t,$$

so generally

# Factor Portfolios – From Characteristics to Rankings

Let

$$x_{i,t} \in \mathbb{R}$$

denote a characteristic used to construct a factor portfolio.

For example,

Book-to-Market  $\longrightarrow$  Value

Past Returns  $\longrightarrow$  Momentum.

At each date  $t$ , stocks are ranked according to  $x_{i,t}$ .

**High-Ranked Group**

$$\mathcal{H}_t$$

Stocks with high values of  $x_{i,t}$ .

**Low-Ranked Group**

$$\mathcal{L}_t$$

Stocks with low values of  $x_{i,t}$ .

# Factor Portfolios – Long/Short Construction

A simple factor portfolio goes

Long High-Ranked Stocks – Short Low-Ranked Stocks

For equal-weighted groups,

$$R_{F,t+1} = \frac{1}{N_H} \sum_{i \in \mathcal{H}_t} R_{i,t+1} - \frac{1}{N_L} \sum_{i \in \mathcal{L}_t} R_{i,t+1}$$

with

$$N_H = |\mathcal{H}_t|, \quad N_L = |\mathcal{L}_t|.$$

# Factor Portfolios – Interpreting the Factor Return

Define the returns of the high- and low-ranked portfolios as

$$R_{H,t+1} = \frac{1}{N_H} \sum_{i \in \mathcal{H}_t} R_{i,t+1},$$

$$R_{L,t+1} = \frac{1}{N_L} \sum_{i \in \mathcal{L}_t} R_{i,t+1}.$$

Therefore,

$$R_{F,t+1} = R_{H,t+1} - R_{L,t+1}$$

## Interpretation

The factor return measures the return spread between stocks with high and low values of the underlying characteristic.

# Factor Investing – Economic Interpretation

Why might a factor earn a persistent premium?

## ① Risk-Based Explanation

Investors are compensated for bearing systematic risk.

## ② Behavioural Explanation

Persistent investor biases can generate predictable mispricing.

## ③ Institutional Explanation

Constraints, mandates, or market structure can create persistent return patterns.

## Empirical Discovery

Some factors are first discovered as statistical patterns in the data (anomalies). The next step is to determine whether the pattern has a plausible economic explanation or is simply the result of data mining.

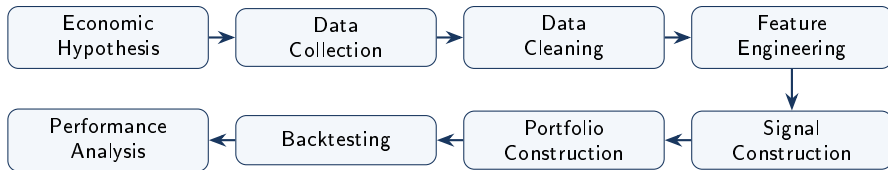
## Important

Statistical evidence of a factor premium does not automatically imply that the factor is economically meaningful, persistent, or investable.

# The Quantitative Research Workflow

# The Quantitative Research Workflow

A quantitative research process follows a sequence of explicit, testable transformations:



## Research Principle

A backtest is the **output of a research process**, not the starting point of the research process.

# The Quantitative Research Workflow – The Investment Hypothesis

Every quantitative strategy should begin with an **explicit and testable investment hypothesis**.

A hypothesis should specify:

- Economic mechanism
- Expected direction
- Investment universe
- Investment horizon
- Expected source of return
- Conditions for failure

## Example – Momentum

Stocks with strong recent performance may continue to outperform stocks with weak recent performance over the next investment horizon.

The economic idea must ultimately become an implementable rule:

Hypothesis → Measurable Variable → Portfolio Rule

# The Quantitative Research Workflow – Data and Feature Engineering

At date  $t$ , let the feature matrix be

$$\mathbf{X}_t = \begin{pmatrix} x_{1,1,t} & \cdots & x_{1,K,t} \\ \vdots & \ddots & \vdots \\ x_{N_t,1,t} & \cdots & x_{N_t,K,t} \end{pmatrix} \in \mathbb{R}^{N_t \times K}$$

where:

$N_t$  = number of assets,       $K$  = number of features.

Feature engineering transforms raw financial data into variables that can be used to construct signals or predictive models.

- Returns
- Growth rates
- Accounting ratios
- Valuation ratios
- Volatility
- Analyst revisions

# The Quantitative Research Workflow – Signal Construction

The feature matrix

$$X_t \in \mathbb{R}^{N_t \times K}$$

is transformed into a vector of investment signals:

$$s_t = f(X_t) \in \mathbb{R}^{N_t}$$

where  $s_{i,t}$  measures the attractiveness of asset  $i$  at date  $t$ .

For example, a linear model may produce

$$s_{i,t} = \beta_0 + \sum_{k=1}^K \beta_k x_{i,k,t}.$$

The signal is then mapped into portfolio weights:

$$w_t = g(s_t) \in \mathbb{R}^{N_t}$$

**Signal  $\neq$  Portfolio:**  $s_t$  measures attractiveness;  $w_t$  determines actual positions.

# The Quantitative Research Workflow – Portfolio Construction

Portfolio construction (PC) maps the signal vector into portfolio weights:

$$w_t = g(s_t)$$

subject to the investor's objectives and constraints.

## Examples of portfolio construction methods:

- Equal weighting
- Rank-based weighting
- Volatility scaling
- Long/short construction
- Mean-variance optimisation
- Risk budgeting
- Factor neutralisation
- Constrained optimisation

## Examples of portfolio constraints:

$$\sum_{i=1}^{N_t} w_{i,t} = 1$$

$$\sum_{i=1}^{N_t} |w_{i,t}| \leq L$$

**Key idea:** PC determines how a research signal becomes an actual investment position.

# The Quantitative Research Workflow – Backtesting

A backtest simulates how a strategy **would have performed historically** using only information available at each date.

Weights determined at time  $t$  are applied to subsequent returns:

$$R_{p,t+1} = w_t^\top R_{t+1}$$

where

$$w_t, R_{t+1} \in \mathbb{R}^{N_t}, \quad R_{p,t+1} \in \mathbb{R}.$$

Starting from initial wealth  $V_0$ , wealth evolves as

$$V_T = V_0 \prod_{t=0}^{T-1} (1 + R_{p,t+1}),$$

so cumulative return is

$$R_{\text{cum},T} = \frac{V_T}{V_0} - 1.$$

# Research Reproducibility

# What is research reproducibility?

Research reproducibility means that the same research process can be executed again to obtain the same results.

A reproducible research process specifies

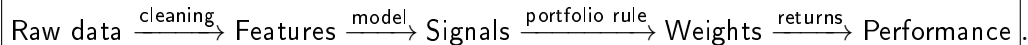
- The data sources.
- The data extraction date.
- The data cleaning rules.
- The feature definitions.
- The model specification.
- The portfolio construction rules.
- The transaction cost assumptions.
- The software environment.
- The random seeds, when relevant.

## Research principle

A result that cannot be reproduced is difficult to verify, maintain or trust.

# Reproducibility and the quantitative workflow

A reproducible strategy can be represented as



Each transformation should be explicit.

- Which data were used?
- Which observations were removed?
- Which parameters were estimated?
- Which dates were used for estimation?
- Which weights were applied?
- Which costs were deducted?

## Important

Reproducibility is not only a software concern. It is also a statistical and financial requirement.

# Research reproducibility: production perspective

In a professional quantitative research environment, reproducibility also requires a reliable software workflow.

## Research tools

- Python.
- NumPy.
- pandas.
- CvxPy.
- scikit-learn.
- Matplotlib.

## Production concepts

- Object-oriented programming.
- Git and GitHub.
- Docker.
- S3.
- EC2.

## Course objective

Introduce Python progressively, while developing the production-level perspective required to turn research into a reliable investment process.

# Summary

# Summary – Chapter 1 – Key Takeaways I

## 1. Quantitative Investing

Explicit rules transform information into signals, portfolios and investment decisions.

## 2. Investment Approaches

**Active vs. passive:** investment objective.

**Discretionary vs. systematic:** decision process.

## 3. Alpha, Beta and Risk Premia

Returns can be analysed through systematic factor exposures, factor premia, alpha and residual risk.

## 4. Portfolio Exposure

Long-only and long/short portfolios differ in constraints, net exposure, gross exposure and neutrality.

# Summary – Chapter 1 – Key Takeaways II

## 5. Cross-Sectional vs. Time-Series

Cross-sectional strategies compare assets **at the same date**.

Time-series strategies exploit an asset's **own history**.

## 6. Factor Investing

Factor strategies target systematic characteristics associated with differences in expected returns or risk.

## 7. Research Workflow

Hypothesis → Data → Signal → Portfolio → Backtest

## 8. Research Discipline

Point-in-time information, reproducibility and robust evaluation are essential for credible research.

## Chapter 2

# Financial Data Engineering

# Chapter 2 – Agenda

## Financial Data

- ➊ Financial Data Sources
- ➋ Prices, Returns and Corporate Actions
- ➌ Fundamental, Macroeconomic and Alternative Data
- ➍ Point-in-Time Data
- ➎ Research Data Architecture

## Building Reliable Research Datasets

- ➏ Merging Heterogeneous Datasets
- ➐ Missing Data
- ➑ Data Validation and Versioning
- ➒ Backtesting Biases
- ➓ Chapter Summary

## Main Objective

Learn how to transform heterogeneous raw financial data into a **clean, point-in-time, reproducible and backtest-ready dataset**.

# Financial Data Sources

# Financial Data Sources – The Data Universe

Quantitative equity strategies can combine several families of data.

## Market Data

- Prices and returns
- Volume and liquidity
- Bid–ask spreads
- Market capitalisation

## Fundamental Data

- Income statements
- Balance sheets
- Cash-flow statements
- Analyst estimates

**Key idea:** Datasets differ in frequency, identifiers, publication mechanisms and availability dates, making their correct alignment a central part of quantitative research.

## Macroeconomic Data

- GDP and inflation
- Interest rates
- Employment
- Business surveys

## Alternative Data

- News and textual data
- Earnings-call transcripts
- Web and transaction data
- Satellite / geolocation data

# Financial Data Sources – Structured vs. Unstructured Data

Financial datasets also differ in their **representation**.

## Structured Data

- Prices
- Accounting variables
- Analyst forecasts
- Macroeconomic releases
- Security master data

Typically represented as tables:

Asset  $\times$  Date  $\times$  Variable.

## Unstructured Data

- News articles
- Earnings-call transcripts
- Regulatory filings
- Images
- Audio

Requires additional processing before quantitative modelling.

**Raw Data  $\rightarrow$  Structured Representation  $\rightarrow$  Features  $\rightarrow$  Signals**

# Financial Data Sources – Frequency and Granularity

Financial datasets are observed at very different frequencies.

| Data               | Typical Frequency       | Example       |
|--------------------|-------------------------|---------------|
| Market prices      | Tick / intraday / daily | Stock price   |
| Trading volume     | Intraday / daily        | Shares traded |
| Fundamentals       | Quarterly / annual      | Earnings      |
| Analyst estimates  | Event-driven / daily    | EPS forecast  |
| Macroeconomic data | Monthly / quarterly     | CPI, GDP      |
| Textual data       | Event-driven            | News, filings |

A research dataset must therefore align observations with different **frequencies**, **timestamps** and **publication schedules**.

## Point-in-Time Principle

The relevant question is not only “What value corresponds to date  $t$ ?”, but “**What information was actually known to the investor at date  $t$ ?**”

# Prices, Returns and Corporate Actions

## Prices and Returns – Price Data

For asset  $i$  at date  $t$ , let

$$P_{i,t} \in \mathbb{R}_+$$

denote its observed market price.

Common market-price fields include:

- Open price
- High price
- Low price
- Close price
- Trading volume
- Adjusted close

For daily equity research, the closing price is often used as the reference price:

$$P_{i,t} = P_{i,t}^{Close}.$$

### Important

Raw prices cannot always be compared directly through time because corporate actions can mechanically change the quoted share price.

## Prices and Returns – Simple Returns

The simple return of asset  $i$  between  $t - 1$  and  $t$  is

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

where

$$P_{i,t-1}, P_{i,t} \in \mathbb{R}_+, \quad R_{i,t} \in \mathbb{R}.$$

Equivalently,

$$P_{i,t} = P_{i,t-1}(1 + R_{i,t})$$

and over several periods,

$$\frac{P_{i,T}}{P_{i,0}} = \prod_{t=1}^T (1 + R_{i,t})$$

when prices are comparable through time and no distributions are omitted.

# Corporate Actions – Why Raw Prices Can Be Misleading

Corporate actions can modify the observed stock price without representing an economic loss or gain for the shareholder.

## Distributions

- Cash dividends
- Special dividends
- Capital distributions

## Share Structure

- Stock splits
- Reverse splits

## Corporate Events

- Spin-offs
- Rights issues
- Mergers and acquisitions

## Example

A 2-for-1 stock split approximately halves the quoted share price, but does not halve the shareholder's wealth: the investor now owns twice as many shares.

## Corporate Actions – Dividends and Total Returns

Suppose asset  $i$  pays a cash dividend  $D_{i,t}$  between  $t - 1$  and  $t$ .

The price return is

$$R_{i,t}^{Price} = \frac{P_{i,t}}{P_{i,t-1}} - 1.$$

The shareholder's return must also account for the dividend:

$$R_{i,t}^{Total} = \frac{P_{i,t} + D_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

or equivalently,

$$R_{i,t}^{Total} = \frac{P_{i,t} + D_{i,t}}{P_{i,t-1}} - 1.$$

### Research Implication

Using price returns instead of total returns can systematically underestimate the performance of dividend-paying stocks and distort cross-sectional comparisons.

# Corporate Actions – Stock Splits

Suppose a stock undergoes a  $q$ -for-1 split.

**Notation:**  $N_t$  = number of shares held,  $P_t$  = share price,  $q$  = split ratio,  $t^- / t^+$  = immediately before / after the split.

Immediately before the split,

$$\text{Wealth}_{t^-} = N_{t^-} P_{t^-}.$$

After the split,

$$N_{t^+} = qN_{t^-}, \quad P_{t^+} \approx \frac{P_{t^-}}{q}.$$

Hence,

$$N_{t^+} P_{t^+} \approx N_{t^-} P_{t^-}$$

so shareholder wealth is approximately unchanged.

However, using the two raw prices directly would give

$$\frac{P_{t^+}}{P_{t^-}} - 1 \approx \frac{1}{q} - 1$$

which is not an economically meaningful investment return.

**Implication:** Prices or returns must be adjusted for stock splits before being used in quantitative research.

# Adjusted Prices and Total-Return Series

Data providers often construct **adjusted price series** to account for corporate actions.

The objective is to obtain a price-like series  $\tilde{P}_{i,t}$  such that

$$R_{i,t}^{Total} = \frac{\tilde{P}_{i,t}}{\tilde{P}_{i,t-1}} - 1$$

captures the economic return to the investor.

Adjustments may incorporate:

- stock splits,
- cash and special dividends,
- rights distributions,
- spin-offs and other corporate actions.

## Practical Rule

Never assume that a field called price, close, adjusted close or return has the same definition across data providers. Always verify the data documentation.